THE COHERATOR DQ SHIFTS BOUNDEDNESS BY A UNIFORM AMOUNT DEPENDING ONLY ON X

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof runs the induction principle with the affine case as base, controls the three pieces of the distinguished square by their individual bounds, and absorbs them into a single N via the finite cohomological dimension of the Rj_* functors

Editorial note. The label lemma-boundedness-better-coherator names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the coherator DQ_X of lemma-better-coherator and its affine description RQ_X = DQ_X as the quasi-coherent sheaf attached to RGamma(X, K) (lemma-affine-coherator of the same chapter)
- the induction principle for properties of quasi-compact quasi-separated algebraic spaces (lemma-induction-principle of the same chapter)
- elementary distinguished squares (U subset W, f : V -> W) with quasi-compact quasi-separated W, quasi-compact open U and affine V, and the Mayer-Vietoris distinguished triangle of remark-explain-consequence
- finite cohomological dimension on D_QCoh of the functors Rj_* for the thickenings j of the square (lemma-quasi-coherence-direct-image of the same chapter)

Source reference. Original reference 08D9: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2441–2441. Prerequisite pointer invoked at proof line(s) 4807; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CR0: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4654–4654. Prerequisite pointer invoked at proof line(s) 4790; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08DR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1563–1563. Prerequisite pointer invoked at proof line(s) 4801; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 08D5: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 790–790. Prerequisite pointer invoked at proof line(s) 4823; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CR2: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4769–4769. Prerequisite pointer invoked at proof line(s) 4829; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. The functor DQ_X of Lemma 0CR0 has the following boundedness property: there exists an integer $N = N(X)$ such that, if K in $D(\mathcal{O}_X)$ with $H^i(U, K) = 0$ for U affine étale over X and $i \notin [a, b]$, then the cohomology sheaves $H^i(DQ_X(K))$ are zero for $i \notin [a, b + N]$.*

Proof. We will prove this using the induction principle of Lemma 08DR.

If X is affine, then the lemma is true with $N = 0$ because then $RQ_X = DQ_X$ is given by taking the complex of quasi-coherent sheaves associated to $R\Gamma(X, K)$. See Lemma 08D9.

Let $(U \subset W, f : V \to W)$ be an elementary distinguished square with W quasi-compact and quasi-separated, $U \subset W$ quasi-compact open, V affine such that the lemma holds for U, V, and $U \times_W V$. Say with integers $N(U)$, $N(V)$, and $N(U \times_W V)$. Now suppose K is in $D(\mathcal{O}_X)$ with $H^i(W, K) = 0$ for all affine W étale over X and all $i \notin [a, b]$. Then $K|_U$, $K|_V$, $K|_{U \times_W V}$ have the same property. Hence we see that $RQ_U(K|_U)$ and $RQ_V(K|_V)$ and $RQ_{U \cap V}(K|_{U \times_W V})$ have vanishing cohomology sheaves outside the interval $[a, b + \max(N(U), N(V), N(U \times_W V))$. Since the functors $Rj_{U,*}$, $Rj_{V,*}$, $Rj_{U \times_W V,*}$ have finite cohomological dimension on D_{QCoh} by Lemma 08D5 we see that there exists an N such that $Rj_{U,*}DQ_U(K|_U)$, $Rj_{V,*}DQ_V(K|_V)$, and $Rj_{U \cap V,*}DQ_{U \times_W V}(K|_{U \times_W V})$ have vanishing cohomology sheaves outside the interval $[a, b + N]$. Then finally we conclude by the distinguished triangle of Remark 0CR2. □